

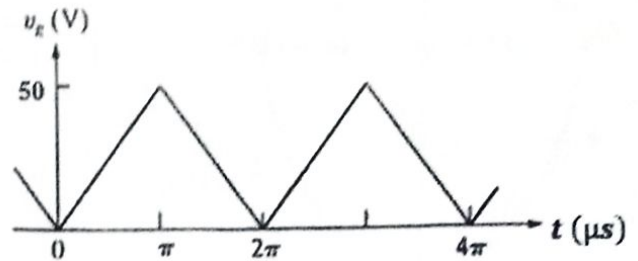
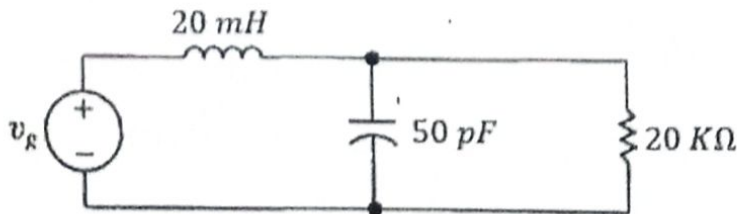
EEC-201 NETWORK THEORY

Time: 3 Hrs.

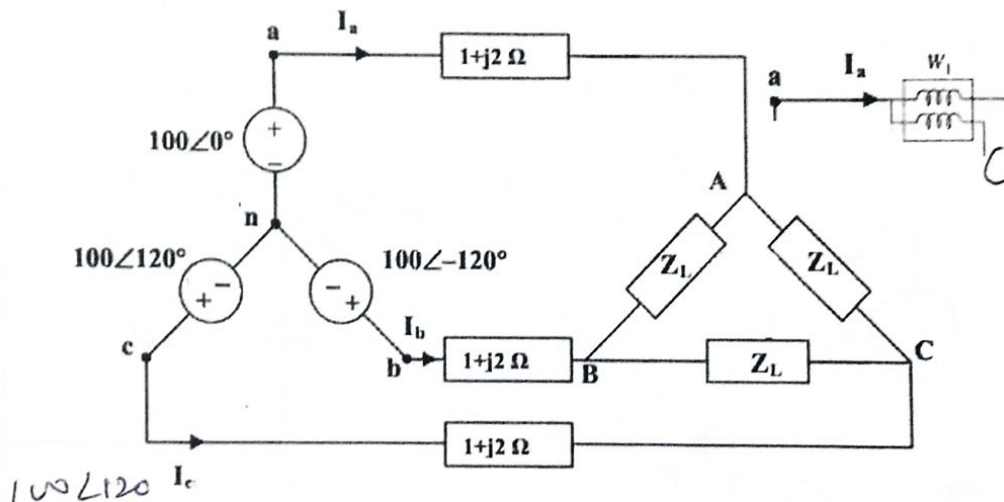
Max. Marks: 90

Answer all questions. If an easier way exists in solving the problem, then not to a complex way.

1. (a) Find the network function of the circuit shown in figure (a) where v_g is the input and the output v_o is the voltage across the resistance. If v_g is a triangular waveform, Estimate the average power delivered to the resistor R when the circuit is in steady-state operation. (Expand v_g up to 3rd harmonic term). [12]

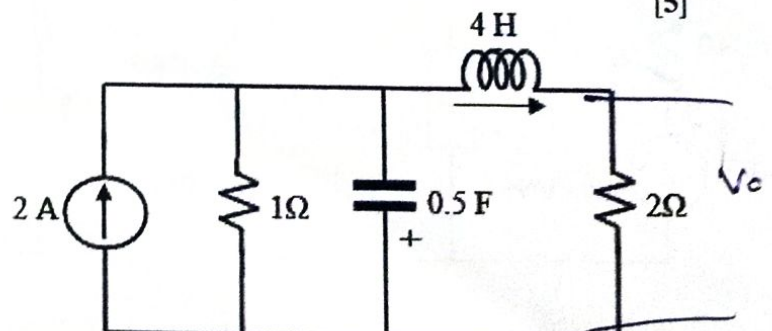
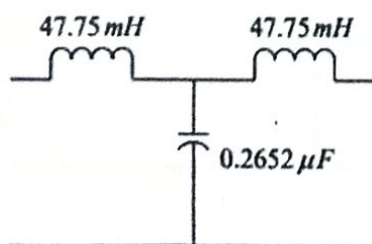


- (b) What is the gain cross over frequency of the network function? What is the phase at this frequency? [4]
2. A balanced 3-phase system is shown in figure where $Z_L = 12 + j12\Omega$.
- (a) Draw a single-phase equivalent of the circuit of the system.
- (b) Determine the line currents.
- (c) Determine the wattmeter reading if the current coil is connected in line a and the pressure (voltage) coil is connected between the points a and C. [11]



- (b) The two sequence components and the corresponding phase voltage of a 3-ph system are $V_{a0} = 1\angle -60^\circ V$; $V_{a1} = 2\angle 0^\circ V$; and $V_a = 3\angle 0^\circ V$. Determine the other phase voltages' [4]

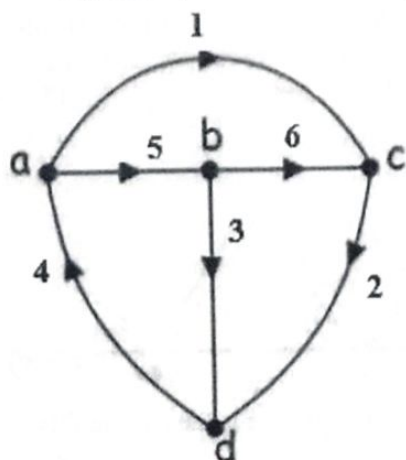
3. (a) Find the characteristic impedance of the two-port network shown in Figure (a). What is the propagation constant of the circuit at $f=3000$ Hz. [5]



- (b) For the network shown in figure (b), obtain a state variable representation with A matrix diagonal. Take the polarity of the capacitor voltage and the direction of the inductor current as shown in figure. [11]

4. (a) Prove that all the branch voltages of a circuit can be represented in terms of the twig voltages as independent voltages.

Represent all the branch voltages of the graph shown in figure (a) in terms of the twig voltages. Take the branches 1, 2 and 3 as the twigs. [8]



```
x0 = ([0; .0]);
ts = [0 20];
[t,x] = ode23('ete1',ts,x0);
[m,n] = size(x);
x(m,:)

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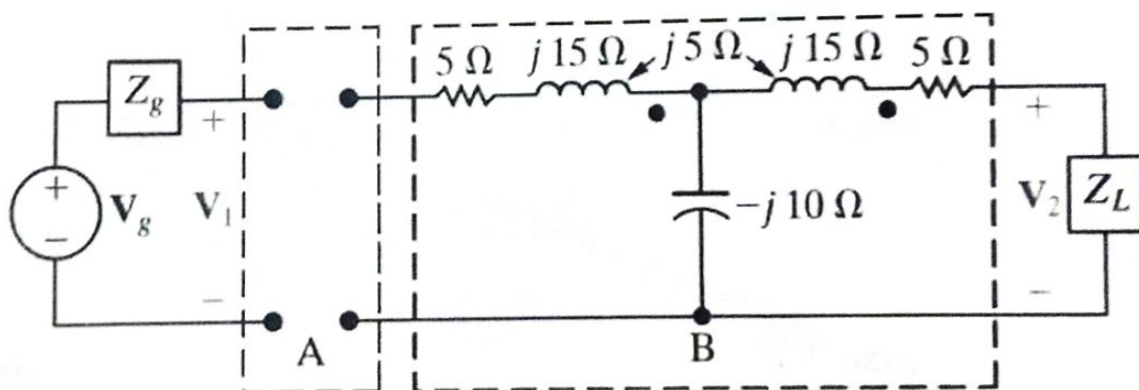
ete1.m

```
function c = ete1(t,x);
n = 4.2;
d = [1 4 0];
so = tf(n,d);
s = feedback(so,1);
[n1 d1] = tfdata(s,'v');
[A,B,C,D] = tf2ss(n1,d1);
u = 1;
c = A*x + B*u;

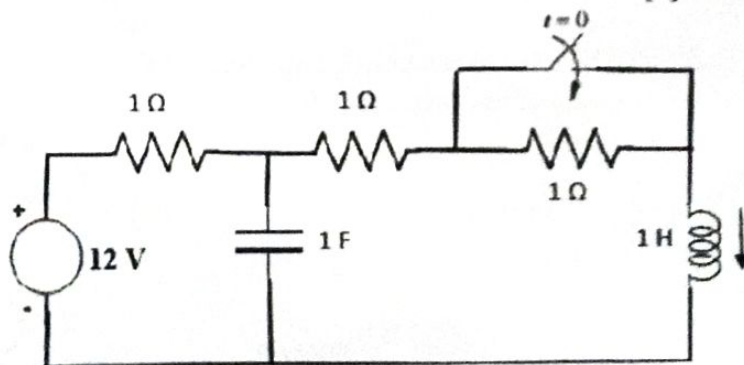
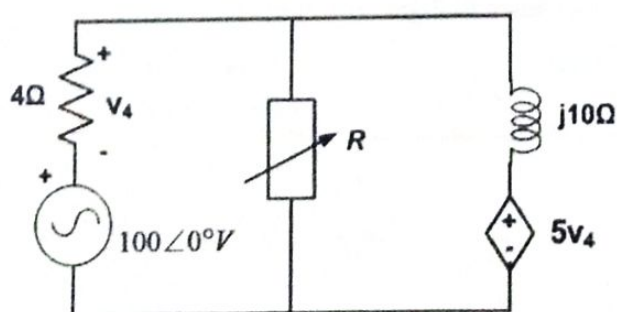
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- (b) In figure (b), which shows the MATLAB commands and function, *tfdata* gives the numerator and denominator which are the input arguments of the command *tf2ss*. Determine the output displayed. What are the eigenvalues of A matrix? Roughly sketch the time history of x_1 (Assuming that x_1 is y and x_2 is $\dot{y}$) showing all the significant points. Explain how did you get the correct time variation. [9]

5. The network A and B in figure are reciprocal and symmetrical. For network A, the transmission parameters A' and B' are 5 and 24Ω respectively. Find V_2 when $V_g = 75 \angle 0^\circ V$, $Z_g = 1 \angle 0^\circ V$, and $Z_L = 10 \angle 0^\circ V$. [11]



6. (a) In the figure (a) shown, determine the value of R for maximum power transfer. Also determine the maximum value of power. [7]



- (b) The switch shown in figure (b) is closed at $t = 0$. Determine the current in the inductor for $t > 0$. [8]